EXPERIMENT 1

AIM:

To design and develop a simple personal portfolio website using basic HTML, CSS, and JavaScript that displays personal details, skills, projects, and contact information

TASKS:

. Create a project folder named personal portfolio.

. Create index.html, style.css, and script.js files.

. Design sections for About, Skills, Projects, and Contact.

. Apply basic styling using CSS.

TECH STACK USED:

HTML, CSS, JavaScript, VS Code (Editor), Web Browser (Chrome)

CODE:

```
JS script.js X
function welcome(): void
experiment_1 > JS script.js > welcome
function welcome() {
    alert("Thank you for visiting!");
}

function submitForm(e) {
    e.preventDefault();
    document.getElementById("msg").innerText = "Message submitted successfully";
}
```

index.html

experiment_1 > index.html > html > body > section#skills > h2

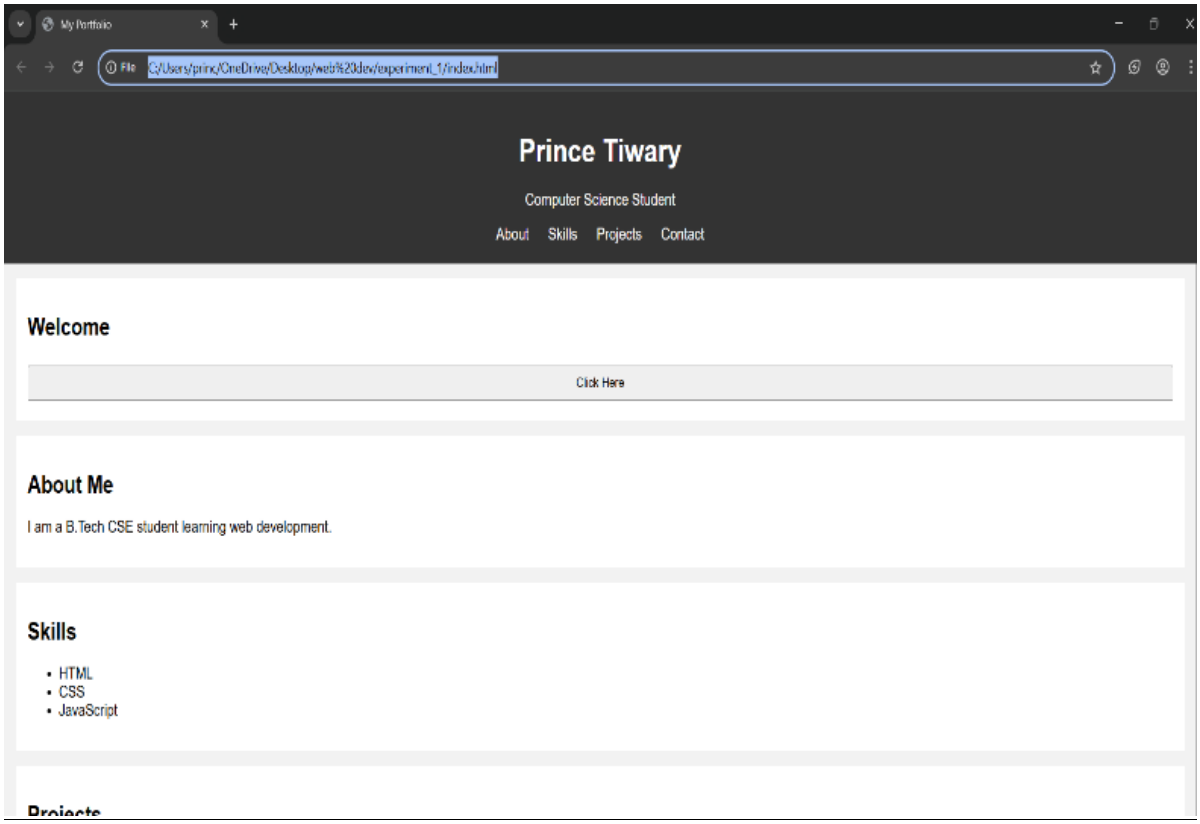
```html
<!DOCTYPE html>
<html>
<head>
    <title>My Portfolio</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<header>
    <h1>Prince Tiwary </h1>
    <p>Computer Science Student</p>

    <nav>
        <a href="#about">About</a>
        <a href="#skills">Skills</a>
        <a href="#projects">Projects</a>
        <a href="#contact">Contact</a>
    </nav>
</header>

<section>
    <h2>Welcome</h2>
    <button onclick="welcome()">Click Here</button>
</section>

<section id="about">
    <h2>About Me</h2>
    <p>I am a B.Tech CSE student learning web development.</p>
</section>

<section id="skills">
    <h2>Skills</h2>
    <ul>
        <li>HTML</li>
        <li>CSS</li>
        <li>JavaScript</li>
    </ul>
</section>

<section id="projects">
    <h2>Projects</h2>
```

style.css

experiment_1 > style.css > body

```css
body {
    font-family: Arial;
    background: #f2f2f2;
    margin: 0;
}

header {
    background: #333;
    color: white;
    text-align: center;
    padding: 20px;
}

nav a {
    color: white;
    margin: 10px;
    text-decoration: none;
}

section {
    background: white;
    margin: 15px;
    padding: 15px;
}

div {
    border: 1px solid #ccc;
    padding: 10px;
    margin-top: 10px;
}

input, textarea, button {
    margin: 5px 0;
    padding: 8px;
    width: 100%;
}

footer {
    background: #333;
    color: white;
    text-align: center;
    padding: 10px;
}
```

OUTPUT:


My Portfolio
File
C:/Users/princ/OneDrive/Desktop/web%20dev/experiment_1/index.html
Prince Tiwary
Computer Science Student
About Skills Projects Contact
Welcome
Click Here
About Me
I am a B.Tech CSE student learning web development.
Skills
HTML
CSS
JavaScript

EXPERIMENT 2

AIM:

To design and develop a Company Landing Page using HTML and CSS that includes a hero section, about section, services, testimonials, and a contact form.

TASKS

. Create a project folder named **company_landing_page**.

. Create **index.html** and **style.css** files inside the folder.

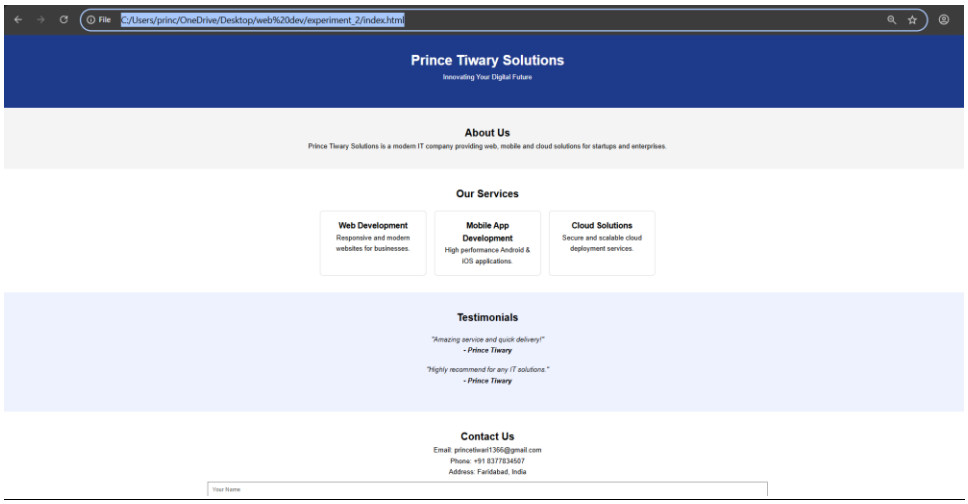
. Add header comments in HTML (project title, name, date).

. Apply CSS styling for layout, spacing, colors, fonts, and alignment using box model properties.

TECK STACK USED

HTML,CSS

OUTPUT:



CODE:

index.html

experiment_2 > index.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>PRINCE TIWARY SOLUTIONS</title>
    <link rel="stylesheet" href="style.css">
</head>

<body>

<header class="hero">
    <h1>Prince Tiwary Solutions</h1>
    <p>Innovating Your Digital Future</p>
</header>

<section id="about">
    <h2>About Us</h2>
    <p>
        Prince Tiwary Solutions  is a modern IT company providing web,
        mobile and cloud solutions for startups and enterprises.
    </p>
</section>

<section id="services">
    <h2>Our Services</h2>

    <div class="service-container">
        <div class="card">
            <h3>Web Development</h3>
            <p>Responsive and modern websites for businesses.</p>
        </div>

        <div class="card">
            <h3>Mobile App Development</h3>
            <p>High performance Android & iOS applications.</p>
        </div>

        <div class="card">
            <h3>Cloud Solutions</h3>
            <p>Secure and scalable cloud deployment services.</p>
        </div>
    </div>
</section>
```

style.css

experiment_2 > style.css > *

```css
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: Arial, sans-serif;
    line-height: 1.6;
}

.hero {
    background: #1e3a8a;
    color: white;
    text-align: center;
    padding: 60px 20px;
}

section {
    padding: 40px;
    text-align: center;
}

#about {
    background: #f4f4f4;
}

.service-container {
    display: flex;
    justify-content: center;
    gap: 20px;
    margin-top: 20px;
}

.card {
    background: white;
    padding: 20px;
    width: 250px;
    border: 1px solid #ddd;
    border-radius: 8px;
}

#testimonials {
    background: #eef2ff;
```

EXPERIMENT 3

AIM:

To design and develop an **Interactive Quiz Web Application** using HTML, CSS and JavaScript that displays questions, tracks score, shows results and allows restarting the quiz.

TASKS:

. Create a project folder named **interactive_quiz**.

. Capture user answers and move between questions.

. Track and update quiz score dynamically.

. Display final result and feedback message.

. Add **Restart Quiz** feature to reuse the app.

TECH STACK USED:

Html, CSS, JavaScript

CODE:

index.html

experiment_3 > index.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Interactive Quiz</title>
    <link rel="stylesheet" href="style.css">
</head>

<body>

<div class="container">
    <h1>General Knowledge Quiz</h1>
    <p>Test your basic knowledge with this quick quiz.</p>

    <div id="quiz-box">
        <h2 id="question">Question here</h2>
        <div id="options"></div>
        <button id="nextBtn">Next</button>
    </div>

    <div id="result-box" class="hide">
        <h2 id="score"></h2>
        <p id="message"></p>
        <button onclick="restartQuiz()">Restart Quiz</button>
    </div>
</div>

<script src="script.js"></script>
</body>
</html>
```

style.css

experiment_3 > style.css > body

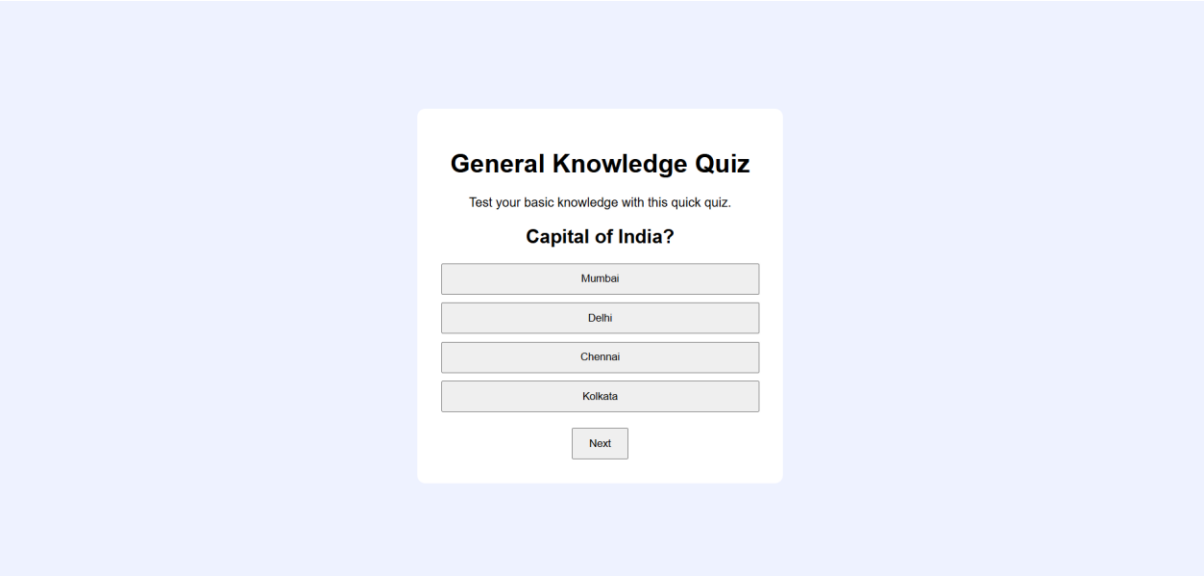
```css
body {
    font-family: Arial;
    background: #eef2ff;
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
}

.container {
    background: white;
    padding: 30px;
    width: 400px;
    text-align: center;
    border-radius: 10px;
}

#options button {
    display: block;
    width: 100%;
    margin: 10px 0;
    padding: 10px;
    cursor: pointer;
}

#nextBtn {
    margin-top: 10px;
    padding: 10px 20px;
}

.hide {
    display: none;
}
```

JS script.js

experiment_3 > JS script.js > ...

```javascript
const questions = [
    {
        question: "Capital of India?",
        options: ["Mumbai", "Delhi", "Chennai", "Kolkata"],
        answer: "Delhi"
    },
    {
        question: "2 + 2 = ?",
        options: ["3", "4", "5", "6"],
        answer: "4"
    },
    {
        question: "HTML stands for?",
        options: ["Hyper Trainer Marking Language", "Hyper Text Markup Language", "High Text Machine Language", "None"],
        answer: "Hyper Text Markup Language"
    }
];

let currentQ = 0;
let score = 0;

const questionEl = document.getElementById("question");
const optionsEl = document.getElementById("options");
const nextBtn = document.getElementById("nextBtn");

function loadQuestion() {
    optionsEl.innerHTML = "";
    let q = questions[currentQ];
    questionEl.textContent = q.question;

    q.options.forEach(option => {
        let btn = document.createElement("button");
        btn.textContent = option;
        btn.onclick = () => selectAnswer(option);
        optionsEl.appendChild(btn);
    });
}

function selectAnswer(option) {
    if (option === questions[currentQ].answer) score++;
    currentQ++;

    if (currentQ < questions.length) {
        loadQuestion();
```

OUTPUT:



EXPERIMENT 4

AIM:

To design and develop a Dynamic To-Do List Web Application that allows users to add, edit, delete, and mark tasks as completed using HTML, CSS, and JavaScript.

TASKS:

. Created project folder dynamic _to do _list

. Designed To-Do interface using HTML

. Implemented dynamic task addition using JavaScript

TECH STACK USED:

HTML,CSS ,JAVASCRIPT

CODE:

index.html

experiment_4 > index.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Dynamic To Do List</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<div class="container">
    <h1>Dynamic To-Do List</h1>
    <p class="desc">Add, edit, delete and complete your daily tasks.</p>

    <div class="input-section">
        <input type="text" id="taskInput" placeholder="Enter a task">
        <button onclick="addTask()">Add Task</button>
    </div>

    <ul id="taskList"></ul>

    <div class="counter">
        Completed: <span id="completed">0</span> |
        Pending: <span id="pending">0</span>
    </div>
</div>

<script src="script.js"></script>
</body>
</html>
```

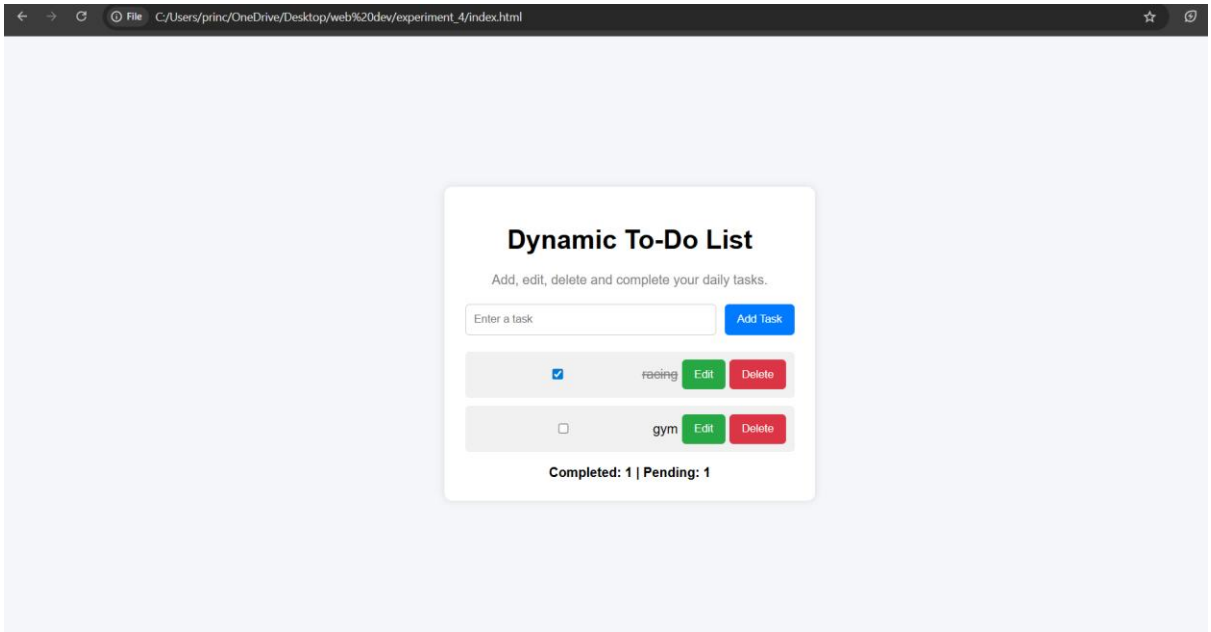
style.css

experiment_4 > style.css > body

```css
body{
    font-family: Arial, sans-serif;
    background: #f4f6f9;
    display:flex;
    justify-content:center;
    align-items:center;
    height:100vh;
}

.container{
    background: white;
    padding:25px;
    width:400px;
    border-radius:10px;
    box-shadow:0 0 10px rgba(0,0,0,0.1);
}

h1{
    text-align:center;
}

.desc{
    text-align:center;
    color: gray;
    margin-bottom:20px;
}

.input-section{
    display:flex;
    gap:10px;
}

input{
    flex:1;
    padding:10px;
    border:1px solid #ccc;
    border-radius:5px;
}

button{
    padding:10px 15px;
    border:none;
    background: #007bff;
```

```
JS script.js
experiment_4 > JS script.js > ...
let taskList = document.getElementById("taskList");
let completedCount = document.getElementById("completed");
let pendingCount = document.getElementById("pending");

function updateCounter(){
    let tasks = document.querySelectorAll("li");
    let completed = document.querySelectorAll(".completed");
    completedCount.textContent = completed.length;
    pendingCount.textContent = tasks.length - completed.length;
}

function addTask(){
    let input = document.getElementById("taskInput");
    let taskText = input.value.trim();

    if(taskText === "") return;

    let li = document.createElement("li");

    let checkbox = document.createElement("input");
    checkbox.type = "checkbox";
    checkbox.onclick = function(){
        li.classList.toggle("completed");
        updateCounter();
    };

    let span = document.createElement("span");
    span.textContent = taskText;

    let btnDiv = document.createElement("div");
    btnDiv.className = "task-btns";

    let editBtn = document.createElement("button");
    editBtn.textContent = "Edit";
    editBtn.onclick = function(){
        let newTask = prompt("Edit task:", span.textContent);
        if(newTask !== null && newTask.trim() !== ""){
            span.textContent = newTask;
        }
    };

    let deleteBtn = document.createElement("button");
    deleteBtn.textContent = "Delete";
    deleteBtn.className = "delete";
```

OUTPUT:



EXPERIMENT 5

AIM:

To design and develop a Simple Blog Web Application using the Flask framework that performs CRUD operations (Create, Read, Update, Delete) without using a database, and demonstrates the use of templates, routing, and basic styling in web development.

TASK :

Project Setup & Environment

Flask App Initialization

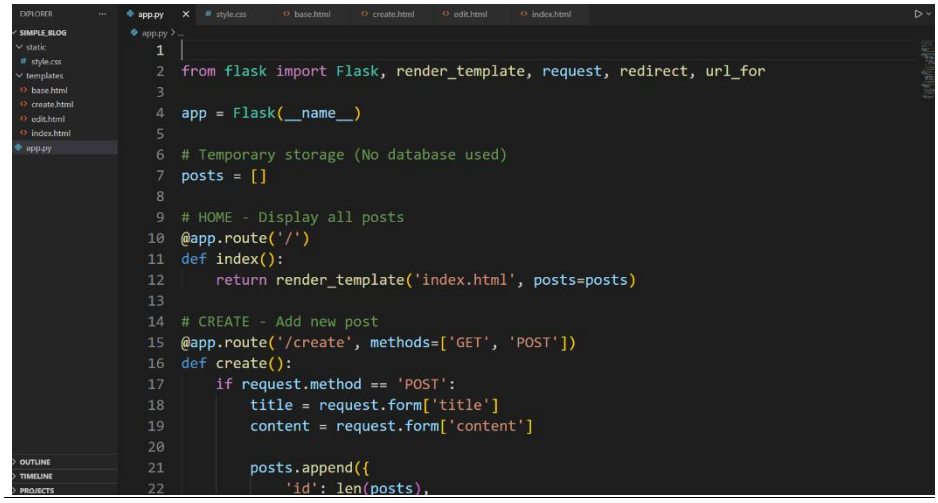
Create Operation

Read Operation

Tech Stack Used:

PYTHON, FLASK, HTML,CSS

CODE:



```python

from flask import Flask, render_template, request, redirect, url_for

app = Flask(__name__)

# Temporary storage (No database used)
posts = []

# HOME - Display all posts
@app.route('/')
def index():
    return render_template('index.html', posts=posts)

# CREATE - Add new post
@app.route('/create', methods=['GET', 'POST'])
def create():
    if request.method == 'POST':
        title = request.form['title']
        content = request.form['content']

        posts.append({
            'id': len(posts),
```

app.py | style.css | base.html | create.html | edit.html | index.html

templates > index.html > ...

```html
{% extends 'base.html' %}

{% block content %}

<h2>📰 Latest Blog Posts</h2>

{% if posts %}
    {% for post in posts %}
    <div class="post">
        <h3>{{ post.title }}</h3>
        <p>{{ post.content }}</p>

        <div class="actions">
            <a href="/edit/{{ post.id }}" class="edit">Edit</a>
            <a href="/delete/{{ post.id }}" class="delete">Delete</a>
        </div>
    </div>
    {% endfor %}
{% else %}
    <p class="empty">No posts yet. Create your first post 🚀</p>
{% endif %}
```

style.css | create.html

templates > create.html > ...

```html
{% extends 'base.html' %}

{% block content %}

<h2>Create New Post ✨</h2>

<form method="POST" class="form-card">
    <input type="text" name="title" pla

    <textarea name="content" placeholde

    <button type="submit">Publish Post</b

</form>

{% endblock %}
```

create.html | edit.html

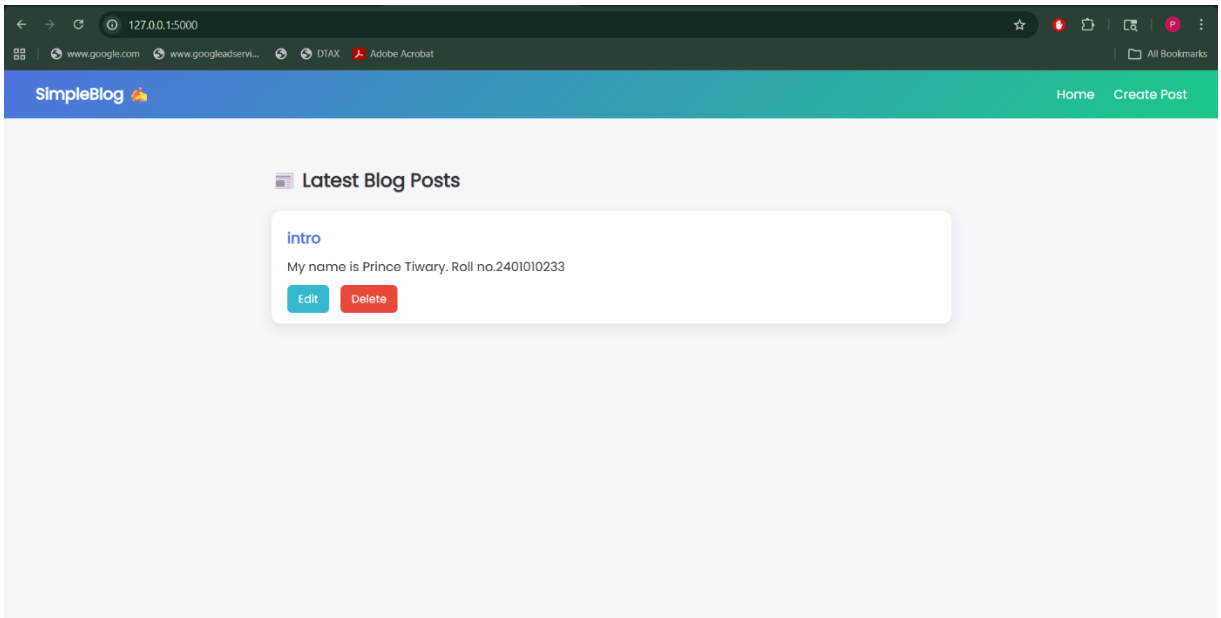
templates > edit.html > ...

```html
{% extends 'base.html' %}

{% block content %}

<h2>Edit Post ✏️</h2>

<form method="POST" class="form-card">
    <input type="text" name="title" val

    <textarea name="content" required>{

    <button type="submit">Update Post</b

</form>

{% endblock %}
```

```html
{% extends 'base.html' %}

{% block content %}

<h2>Create New Post ✨</h2>

<form method="POST" class="form-card">
    <input type="text" name="title" pla

    <textarea name="content" placeholde

    <button type="submit">Publish Post</b
</form>

{% endblock %}
```

```html
{% extends 'base.html' %}

{% block content %}

<h2>Edit Post 🖍</h2>

<form method="POST" class="form-card">
    <input type="text" name="title" val

    <textarea name="content" required>{

    <button type="submit">Update Post</b
</form>

{% endblock %}
```

OUTPUT:



EXPERIMENT 6

AIM:

To develop a Contact Management System using Python and Flask that allows users to perform CRUD operations (Create, Read, Update, Delete) on contact data through a web interface

TASK:

Project Setup & Environment

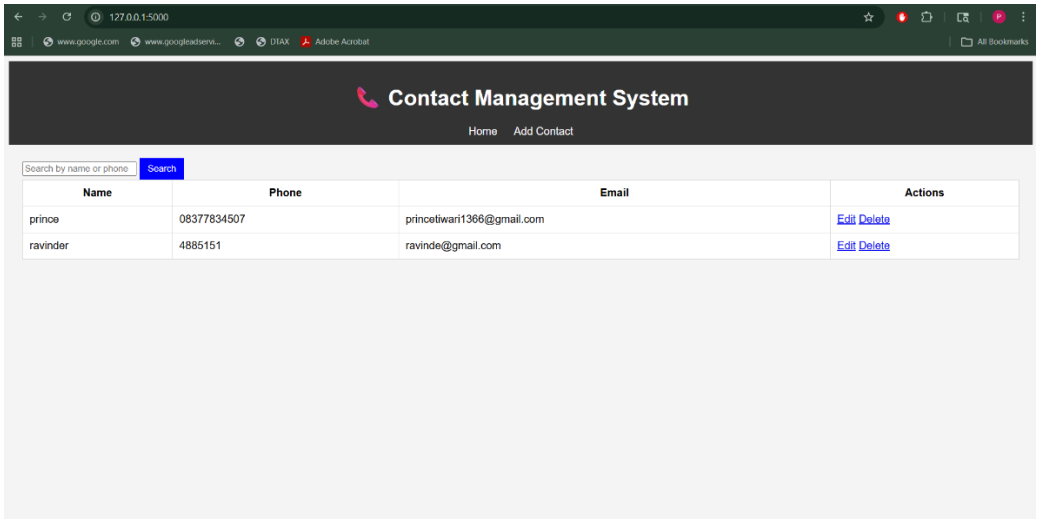
Flask App Initialization

Create Operation (Add Contact)

TECH STACK :

HTML,CSS,PYTHON ,VS CODE

OUTPUT:



CODE:

app.py style.css

app.py > ...

```python

from flask import Flask, render_template, request, redirect, url_for

app = Flask(__name__)

# In-memory storage (list of dictionaries)
contacts = []

# Home Route - Display Contacts
@app.route('/')
def index():
    return render_template('index.html', contacts=contacts)

# Add Contact
@app.route('/add', methods=['GET', 'POST'])
def add_contact():
    if request.method == 'POST':
        name = request.form['name']
        phone = request.form['phone']
        email = request.form['email']

        # Validation
        if name and phone and email:
            contacts.append({
                'id': len(contacts),
                'name': name,
                'phone': phone,
                'email': email
            })

        return redirect(url_for('index'))
```

app.py base.html style.css add_contact.html

templates > base.html > html

```html
<!DOCTYPE html>
<html>
<head>
    <title>Contact Manager</title>
    <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
</head>
<body>

<header>
    <h1>📞 Contact Management System</h1>
    <nav>
        <a href="/">Home</a>
        <a href="/add">Add Contact</a>
    </nav>
</header>

<div class="container">
    {% block content %}{% endblock %}
</div>

</body>
</html>
```

index.html

templates > index.html > ...

```html
{% extends 'base.html' %}

{% block content %}

<form action="/search" method="POST">
    <input type="text" name="query" placeholder="Search by name or phone">
    <button type="submit">Search</button>
</form>

<table>
    <tr>
        <th>Name</th>
        <th>Phone</th>
        <th>Email</th>
        <th>Actions</th>
    </tr>

    {% for contact in contacts %}
    <tr>
        <td>{{ contact.name }}</td>
        <td>{{ contact.phone }}</td>
        <td>{{ contact.email }}</td>
        <td>
            <a href="/edit/{{ contact.id }}">Edit</a>
            <a href="/delete/{{ contact.id }}">Delete</a>
        </td>
    </tr>
    {% endfor %}
</table>

{% endblock %}
```

add_contact.html

templates > add_contact.html > ...

```html
{% extends 'base.html' %}

{% block content %}
<h2>Add Contact</h2>

<form method="POST">
    <input type="text" name="name" placeholder="Na
    <input type="text" name="phone" placeholder="P
    <input type="email" name="email" placeholder='
    <button type="submit">Add</button>
</form>

{% endblock %}
```

edit_contact.html

templates > edit_contact.html > ...

```html
{% extends 'base.html' %}

{% block content %}
<h2>Edit Contact</h2>

<form method="POST">
    <input type="text" name="name" value="{{ cont
    <input type="text" name="phone" value="{{ con
    <input type="email" name="email" value="{{ co
    <button type="submit">Update</button>
</form>

{% endblock %}
```

app.py | style.css

static > style.css > button

```css
body {
    font-family: Arial;
    background: #f4f4f4;
}

header {
    background: #333;
    color: white;
    padding: 10px;
    text-align: center;
}

nav a {
    color: white;
    margin: 10px;
    text-decoration: none;
}

.container {
    padding: 20px;
}

table {
    width: 100%;
    background: white;
    border-collapse: collapse;
}

th, td {
    padding: 10px;
    border: 1px solid #ddd;
}
```

EXPERIMENT 7

AIM:

To design and develop a multi-page E-commerce website using HTML, CSS, and JavaScript that simulates product browsing, shopping cart functionality, and checkout process.

TASK:

Project Setup & Structure

Product Listing Page

Shopping Cart Functionality

Cart Summary & Price Calculation

Enhanced Features

TECH STACK:

Html ,CSS , JavaScript , vs code

CODE:

index.html × cart.html checkout.html style.css script.js

index.html > html

```html
<!DOCTYPE html>
<html>
<head>
    <title>PT Store</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<header class="navbar">
    <h2 class="logo">PT Store</h2>

    <div>
        <input type="text" id="search" placeholder="Search products...">
        <button onclick="searchProduct()">Search</button>
    </div>

    <a href="cart.html" class="cart-btn">🛒 Cart</a>
</header>

<section class="hero">
    <h1>Welcome to PT Store</h1>
    <p>Best Deals on Electronics</p>
```

cart.html | style.css | script.js

cart.html > html

```html
<!DOCTYPE html>
<html>
<head>
    <title>Cart - PT Store</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<header class="navbar">
    <h2 class="logo">PT Store</h2>
    <a href="index.html" class="cart-btn">⬅ Back</a>
</header>

<h2 style="padding:20px;">Your Cart</h2>

<ul id="cart-items"></ul>

<h3 style="padding-left:20px;">Total: ₹<span id="total"

<a href="checkout.html">
    <button class="checkout-btn">Checkout</button>
</a>

<script src="script.js"></script>
</body>
</html>
```

checkout.html

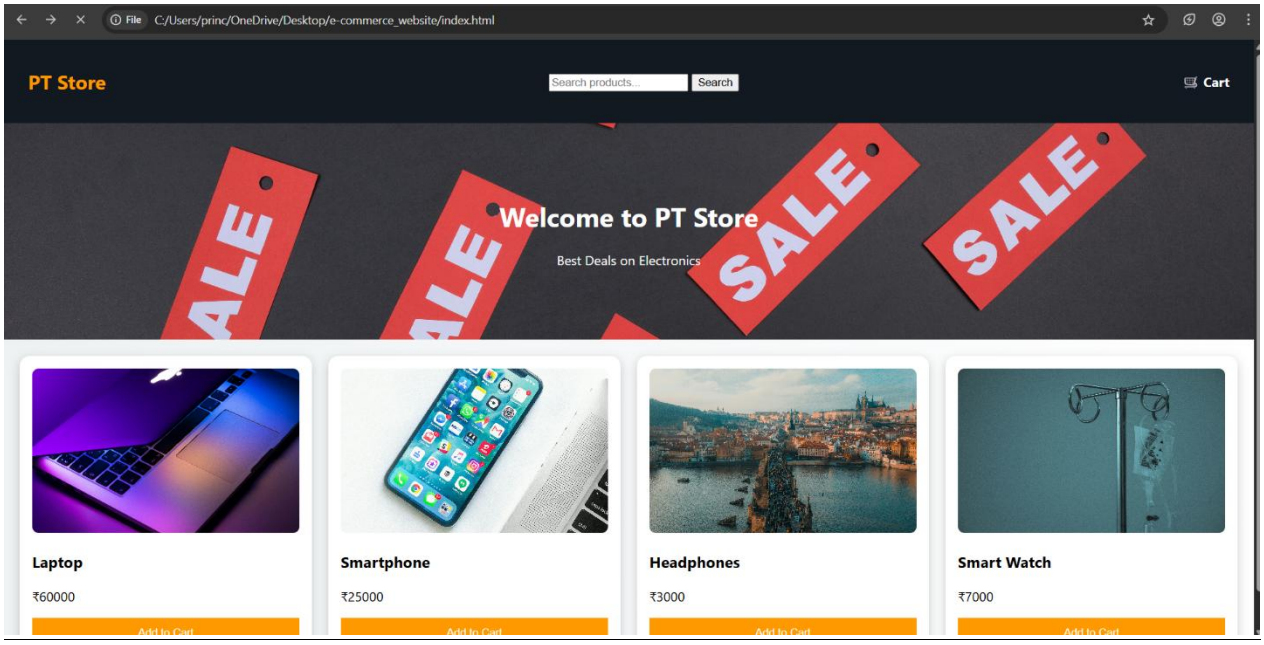
checkout.html > html

```html
<!DOCTYPE html>
<html>
<head>
    <title>Checkout - PT Store</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<header class="navbar">
    <h2 class="logo">PT Store</h2>
</header>

<div class="checkout-box">
    <h2>Checkout</h2>

    <input type="text" id="name" placeholder="Full Na
    <input type="text" id="address" placeholder="Addre

    <button onclick="placeOrder()">Place Order</butto
</div>

<script src="script.js"></script>
</body>
</html>
```

style.css | script.js

style.css > .checkout-box input

```css
body {
    margin: 0;
    font-family: 'Segoe UI', sans-serif;
    background: linear-gradient(to right, #eef2f3, #ffffff);
}

/* NAVBAR */
.navbar {
    background: #131921;
    color: white;
    padding: 15px 30px;
    display: flex;
    justify-content: space-between;
    align-items: center;
}

.logo {
    color: #ff9900;
}

.cart-btn {
    color: white;
    text-decoration: none;
    font-weight: bold;
}

/* HERO */
.hero {
    background: url("https://images.unsplash.com/photo-1607083206968-13611e3d76db") center/cover;
    color: white;
    padding: 70px;
    text-align: center;
```

```
script.js
script.js > placeOrder
const products = [
    {
        id: 1,
        name: "Laptop",
        price: 60000,
        img: "https://images.unsplash.com/photo-1517336714731-489689fd1ca8"
    },
    {
        id: 2,
        name: "Smartphone",
        price: 25000,
        img: "https://images.unsplash.com/photo-1511707171634-5f897ff02aa9"
    },
    {
        id: 3,
        name: "Headphones",
        price: 3000,
        img: "https://images.unsplash.com/photo-1519677100203-a0e668c92439"
    },
    {
        id: 4,
        name: "Smart Watch",
        price: 7000,
        img: "https://images.unsplash.com/photo-1516574187841-cb9cc2ca948b"
    }
];

// LOAD CART FROM STORAGE
let cart = JSON.parse(localStorage.getItem("cart")) || [];

// DISPLAY PRODUCTS
if (document.getElementById("products")) {
```

OUTPUT: